



JABchem



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Past Papers Higher Chemistry

2018 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	80+	66.6%	28.3%
B	67+	55.8%	25.3%
C	54+	45.0%	23.0%
D	47+	39.2%	9.5%
No award	<47	<39.2%	13.9%

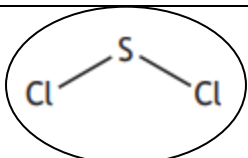
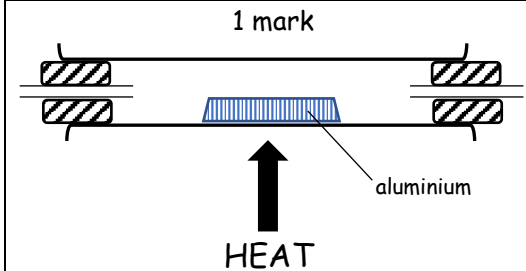
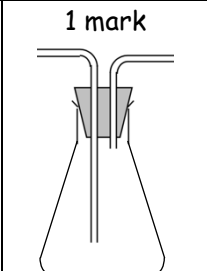
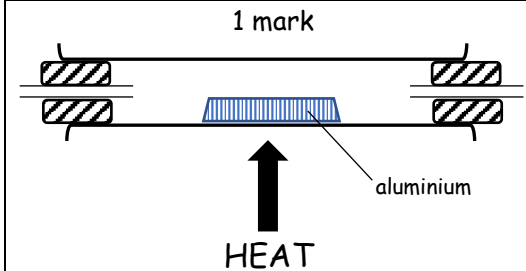
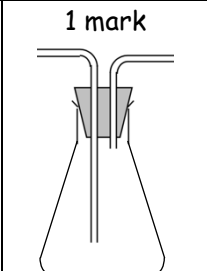
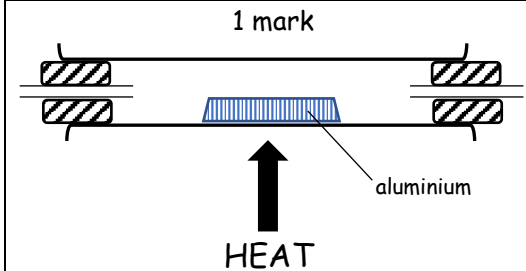
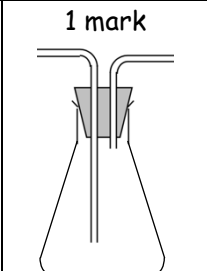
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	12.1 /20	42.9 /80	13.1 /20

2018 Higher Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning								
1	B	70	☒ A Forward reaction has an activation energy of +40kJ mol ⁻¹ but always positive ☒ B Enthalpy change is difference between P and R and downhill means exothermic ☒ C This is enthalpy change for forward reaction from R to P (uphill = endothermic) ☒ D This is the activation energy for the reverse reaction (P to top of hill)								
2	A	91	$\text{Rate} = \frac{1}{t} \therefore t = \frac{1}{\text{rate}} = \frac{1}{100 \text{ s}} = 0.01 \text{ s}^{-1}$								
3	D	70	☒ A Decrease in temp decreases number of collisions with energy greater than E _a ☒ B Activation Energy (E _a) is independent of temperature ☒ C Activation Energy (E _a) is independent of temperature ☒ D No change to activation energy & number of collisions decrease as temp decreases								
4	C	57	$\begin{array}{l} \textcircled{1} \quad \text{Al}^+(\text{g}) \rightarrow \text{Al}^{2+}(\text{g}) + \text{e}^- \quad \Delta H = 1817 \text{ kJ} \\ \textcircled{2} \quad \text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^- \quad \Delta H = 2745 \text{ kJ} \\ \text{Add } \textcircled{1} + \textcircled{2} \quad \text{Al}^+(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + 2\text{e}^- \quad \Delta H = 4562 \text{ kJ} \end{array}$								
5	D	65	☒ A Boron forms covalent network. No molecules to have London Dispersion Forces between ☒ B Neon is a noble gas and contains no covalent bonds ☒ C Sodium is an alkali metal and contains metallic bonds between atoms ☒ D Sulphur has covalent bonds within its S ₈ rings and LDF between molecules.								
6	C	77	<table border="1"> <tr> <td>—O—H</td> <td>$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—OH} \end{array}$</td> <td>$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—O—} \end{array}$</td> <td>$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—} \end{array}$</td> </tr> <tr> <td>hydroxyl group</td> <td>carboxyl group</td> <td>ester group</td> <td>carbonyl group</td> </tr> </table>	—O—H	$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—OH} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—O—} \end{array}$	$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—} \end{array}$	hydroxyl group	carboxyl group	ester group	carbonyl group
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hydroxyl group	carboxyl group	ester group	carbonyl group								
7	A	39	☒ A main chain has 4 carbons, carboxyl functional group and methyls on C ₂ , C ₂ and C ₃ ☒ B side groups incorrectly numbered as C ₂ has two methyl groups attached. ☒ C carbon side groups e.g. methyl groups never appear on carbon number 1 ☒ D longest chain with functional group is four carbons $\therefore$ ends in butanoic acid								
8	D	64	☒ A CH ₃ CH ₂ CH(OH)CH ₂ CH ₃ is pentan-3-ol so cannot be an isomer of pentan-3-ol ☒ B Structure has formula C ₆ H ₁₃ OH so has different formula and not an isomer ☒ C Structure has formula C ₅ H ₉ OH so has different formula and not an isomer ☒ D Structure is pentan-1-ol, has same formula and a different structure								
9	A	69	4-methylpentan-2-ol C ₆ H ₁₄ O $\rightarrow$ 4-methylpentan-2-one C ₆ H ₁₂ O $\therefore$ loss of 2xH atoms ☒ A Loss of 2g per mole would be losing 2xH atoms from 1mol of C ₆ H ₁₄ O ☒ B Gain of 2g per mole would be gaining 2xH atoms on 1mol of C ₆ H ₁₄ O ☒ C Loss of 16g per mole would be losing 1xO atom from 1mol of C ₆ H ₁₄ O ☒ D Gain of 16g per mole would be gaining 1xO atom on 1mol of C ₆ H ₁₄ O								
10	B	93	☒ A All amino acids are necessary for building the different types of proteins ☒ B Essential amino acids are the amino acids humans must acquire in their diet. ☒ C Plants produce the essential amino acids that humans consume in their diet ☒ D All types of amino acids are produced when a protein is hydrolysed								
11	D	43	$\begin{array}{l} n \text{ CO} + (2n+1) \text{ H}_2 \rightarrow n \text{ H}_2\text{O} + \text{C}_n\text{H}_x \\ n \text{ CO} + 2n \text{ H}_2 + \text{H}_2 \rightarrow n \text{ H}_2\text{O} + \text{C}_n\text{H}_x \\ n \text{ CO} + n \text{ H}_2 + n \text{ H}_2 + \text{H}_2 \rightarrow n \text{ H}_2\text{O} + \text{C}_n\text{H}_x \\ n \text{ C} + n \text{ H}_2 + n \text{ H}_2 + \text{H}_2 \rightarrow n \text{ H}_2 + \text{C}_n\text{H}_x \\ n \text{ C} + 2n \text{ H} + 2\text{H} \rightarrow \text{C}_n\text{H}_x \\ n \text{ C} + 2n+2 \text{ H} \rightarrow \text{C}_n\text{H}_x \end{array}$ n Carbon atoms has 2n+2 Hydrogen atoms available: general formula = C_nH_{2n+2}								

12	D	34	1mol Na⁺Cl⁻ contains 1mol Cl⁻ ions ∴ 0.6mol Na⁺Cl⁻ contains 0.6mol Cl⁻ ions 1mol Na⁺Cl⁻ contains 1mol Na⁺ ions ∴ 0.6mol Na⁺Cl⁻ contains 0.6mol Na⁺ ions 1mol (Na⁺)₂SO₄²⁻ contains 1mol SO₄²⁻ ions ∴ 0.2mol (Na⁺)₂SO₄²⁻ contains 0.2mol SO₄²⁻ ions 1mol (Na⁺)₂SO₄²⁻ contains 2mol Na⁺ ions ∴ 0.2mol (Na⁺)₂SO₄²⁻ contains 0.4mol Na⁺ ions
13	A	48	☒ A 0.20g of H₂ ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{0.20}{2} = 0.10\text{mol}$ ∴ largest volume ☒ B 0.44g of CO₂ ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{0.44}{44} = 0.01\text{mol}$ ∴ smallest volume ☒ C 0.60g of Ne ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{0.60}{20} = 0.03\text{mol}$ ☒ D 0.80g of Ar ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{0.80}{2} = 0.02\text{mol}$
14	A	52	$ \begin{array}{ccccccc} 3\text{CuO}_{(s)} & + & 2\text{NH}_{3(g)} & \longrightarrow & 2\text{Cu}_{(s)} & + & \text{N}_{2(g)} & + & 3\text{H}_2\text{O}_{(l)} \\ \text{3mol} & & \text{2mol} & & \text{2mol} & & \text{1mol} & & \text{3mol} \\ \text{negligible volume} & & \text{2vol} & & \text{negligible volume} & & \text{1vol} & & \text{negligible volume} \\ - & & 100\text{cm}^3 & & - & & 50\text{cm}^3 & & - \end{array} $
15	D	43	☒ A Hydrogen gas (H₂) would have no effect as it is neither a reactant nor product ☒ B HCl_(g) would dissolve in water to form acid sending equilibrium to left to remove H⁺ ions ☒ C Cl⁻ ions added so equilibrium would move to left to remove extra Cl⁻ ions ☒ D OH⁻ ions would neutralise H⁺ ions sending equilibrium to right to replace H⁺ ions.
16	C	67	gfm C₄H₉OH = 72 mass = 3.6g no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{3.6}{72} = 0.05\text{mol}$ 0.05mol C₄H₉OH releases 124kJ 1mol C₄H₉OH releases 124kJ × 1/0.05 = -2480kJ mol⁻¹
17	B	75	$\begin{array}{lcl} \text{①} & \text{C}_{(s)} + \text{O}_{2(g)} & \rightarrow \text{CO}_2 & \Delta H = -394\text{kJ} \\ \text{②} & \text{CO}_{(g)} + \frac{1}{2}\text{O}_{2(g)} & \rightarrow \text{CO}_{2(g)} & \Delta H = -283\text{kJ} \\ \text{①} & \text{C}_{(s)} + \text{O}_{2(g)} & \rightarrow \text{CO}_2 & \Delta H = -394\text{kJ} \\ \text{②} \times -1 & \text{CO}_{2(g)} & \rightarrow \text{CO}_{(g)} + \frac{1}{2}\text{O}_{2(g)} & \Delta H = +283\text{kJ} \\ \text{Add ①+②'} & \text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} & \rightarrow \text{CO}_{(g)} & \Delta H = -111\text{kJ} \end{array}$

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Long Qu	Answer	Reasoning																
1a(i)	The attraction an atom/nucleus has for the electrons in a bond/shared electrons	Electronegativity is a measure of the attraction for the electrons in a bond by the nuclei at either end of that bond. Non-metals tend to have higher values of electronegativity and have a higher attraction for the electrons in a bond.																
1a(ii)	One answer from:	<table><tr><td>Increased shielding/ more shielding (from additional shells)</td><td>Covalent radius increases/atom size increases/more shells so attraction of the nucleus/protons for the (outer/shared) electrons decreases</td></tr></table>	Increased shielding/ more shielding (from additional shells)	Covalent radius increases/atom size increases/more shells so attraction of the nucleus/protons for the (outer/shared) electrons decreases														
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1b	Answer to include:	<table><tr><td><u>1 mark</u> (Intermolecular) forces/bonds increase (going down the group)</td><td><u>1 mark</u> London Dispersion Forces are the forces (broken) between the molecules</td><td><u>1 mark</u> The more electrons the stronger the London Dispersion Forces</td></tr></table>	<u>1 mark</u> (Intermolecular) forces/bonds increase (going down the group)	<u>1 mark</u> London Dispersion Forces are the forces (broken) between the molecules	<u>1 mark</u> The more electrons the stronger the London Dispersion Forces													
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2a	Increasing number of protons or increasing nuclear charge	Across a period the same outer shell is being filled by electrons. The number of protons in the nucleus is increasing as is the positive charge in the nucleus. The outer shell of electrons is pulled in closer to the nucleus due to electrostatic attraction making the atom size decrease.																
2b(i)		<table><tr><td>Electronegativity</td><td>Si=1.9</td><td>P=2.2</td><td>S=2.5</td></tr><tr><td>Chlorine</td><td>Cl=3.0</td><td>Cl=3.0</td><td>Cl=3.0</td></tr><tr><td>Difference</td><td>1.1</td><td>0.8</td><td>0.5</td></tr><tr><td>Commentary</td><td>Most polar bonds</td><td>-</td><td>Least polar bonds</td></tr></table>	Electronegativity	Si=1.9	P=2.2	S=2.5	Chlorine	Cl=3.0	Cl=3.0	Cl=3.0	Difference	1.1	0.8	0.5	Commentary	Most polar bonds	-	Least polar bonds
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2b(ii)	Answer to include:	<table><tr><td>1 mark</td><td>Silicon tetrachloride and hexane are both non-polar</td></tr><tr><td>1 mark</td><td>Silicon tetrachloride is non-polar due to its tetrahedral shape (where polarities over molecule cancel out)</td></tr></table>	1 mark	Silicon tetrachloride and hexane are both non-polar	1 mark	Silicon tetrachloride is non-polar due to its tetrahedral shape (where polarities over molecule cancel out)												
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2c(i)	Answer to include:	<table><tr><td>1 mark</td><td>Silicon nitride is a covalent network</td></tr><tr><td>1 mark</td><td>Covalent bonds need to be broken before it will melt</td></tr></table>	1 mark	Silicon nitride is a covalent network	1 mark	Covalent bonds need to be broken before it will melt												
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2c(ii)	17.9	$\text{atom economy} = \frac{140.3}{(3 \times 170.1) + (16 \times 17.0)} \times 100 = \frac{140.3}{510.3 + 272.0} \times 100 = 17.9\%$																
2d(i)	Answer to include:	<table><tr><td> 1 mark</td><td> 1 mark</td></tr></table>	 1 mark	 1 mark														
 1 mark	 1 mark																	
2d(ii)	To supply activation energy	Although the reaction is exothermic, sufficient energy to form the activated complex initially must be supplied for the activation energy barrier to be overcome. Once the reaction gets going the exothermic reaction will provide the heat energy to maintain the reaction.																
3a	water bath or heating mantle or hot plate	A flame-based method of heating should not be used as the reactants and products are volatile and flammable.																
3b	To condense any escaping gases	The reactants and products in the formation of esters can have low enough boiling points to evaporate and escape from the test tube. A cold surface like a small test tube filled with cold water will give a surface for the escaping gases to condense on. (Be careful, condensing gases give off heat which will rapidly heat up the water in the cold test tube)																
3c(i)	water	$\text{benzoic Acid} + \text{methanol} \rightarrow \text{methyl benzoate} + \text{water}$ $\text{C}_6\text{H}_5\text{COOH} + \text{CH}_3\text{OH} \rightarrow \text{C}_6\text{H}_5\text{COOCH}_3 + \text{H}_2\text{O}$																

3c(ii)	Answer showing:	$\text{no. of mol C}_6\text{H}_5\text{COOH} = \frac{\text{mass}}{\text{gfm}} = \frac{5.0}{122} = 0.0410\text{mol}$ $\text{no. of mol CH}_3\text{OH} = \frac{\text{mass}}{\text{gfm}} = \frac{2.5}{32} = 0.0781\text{mol}$ $\text{C}_6\text{H}_5\text{COOH} + \text{CH}_3\text{OH} \rightarrow \text{C}_6\text{H}_5\text{COOCH}_3 + \text{H}_2\text{O}$ 1mol 0.0410mol Available 1mol 0.0410mol Required 0.0410mol CH₃OH required and 0.0781 mol CH₃OH available ∴ CH₃OH in excess and C₆H₅COOH is limiting factor as a result.															
3c(iii)	12.84	500g Benzoic Acid. —————→ £39.80 5g Benzoic acid. —————→ £39.80 × ⁵ / ₅₀₀ = £0.3980 ∴ 3.1g methyl benzoate —————→ £0.3980 100g methyl benzoate —————→ £0.3980 × ¹⁰⁰ / _{3.1} = £12.84															
4a	One diagram from:	$\begin{array}{ccccccc} & \text{H} & \text{O} & \text{H} & \text{H} & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & \text{H} & \text{H} & \end{array}$ pentan-2-one $\begin{array}{ccccccc} & \text{H} & \text{H} & \text{O} & \text{H} & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{H} \\ & & & & & & \\ & \text{H} & \text{H} & & \text{H} & \text{H} & \end{array}$ pentan-3-one $\begin{array}{ccccccc} & \text{H} & \text{H} & \text{O} & \text{H} & & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{H} & \\ & & & & & & \\ & \text{H} & \text{H} & & \text{H} & & \end{array}$ 3-methylbutan-2-one															
4b	One oxidising agent from:	<table><tr><th>Oxidising Agent</th><th>Start Colour</th><th>End Colour</th></tr><tr><td>Acidified Dichromate</td><td>Orange</td><td>Green</td></tr><tr><td>Benedict's/Fehling's Solution</td><td>Blue</td><td>Brick Red (orange)</td></tr><tr><td>Hot copper (II) oxide</td><td>Black</td><td>Brown</td></tr><tr><td>Tollen's Reagent</td><td>(Colourless)</td><td>Silver mirror produced</td></tr></table>	Oxidising Agent	Start Colour	End Colour	Acidified Dichromate	Orange	Green	Benedict's/Fehling's Solution	Blue	Brick Red (orange)	Hot copper (II) oxide	Black	Brown	Tollen's Reagent	(Colourless)	Silver mirror produced
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4c	Permanent dipole to permanent dipole	3-methylbutanal contains a carbonyl C=O bond. There is an electronegativity difference of 1.0 within the bond making the bond polar. The δ+ and δ- charges across the bond are attractive to neighbouring molecules also with a carbonyl C=O group.															
4d	It will oxidise	Aldehydes will oxidise to carboxylic acids which gives food an unpleasant taste known as rancid.															
4e(i)	Two molecules join together with the loss of water/small molecule	<table><tr><td>Condensation</td><td>Two small molecules join together to form a larger molecule with the loss of a small molecules (usually water)</td></tr><tr><td>Hydrolysis</td><td>A larger molecule splits into two smaller molecules with a small molecule (usually water) added across the break point</td></tr></table>	Condensation	Two small molecules join together to form a larger molecule with the loss of a small molecules (usually water)	Hydrolysis	A larger molecule splits into two smaller molecules with a small molecule (usually water) added across the break point											
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4e(ii)	6-methylheptan-2-one	$\begin{array}{ccccccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{O} & \text{H} & & & \\ & & & & & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \text{H} & & \\ & & & & & & & & & & \\ & \text{H} & & \text{H} & \text{H} & \text{H} & & \text{H} & & & \\ & & \text{H} - \text{C} - \text{H} & & & & & & & & \\ & & & & & & & & & & \\ & & \text{H} & & & & & & & & \end{array}$ C=O Carbonyl group between two carbons ∴ -one C=O Carbonyl group on C₂ ∴ -2-one Seven carbons on main chain with functional group ∴ heptan-2-one -CH₃ methyl group on C₆ ∴ 6-methylheptan-2-one															
5	Open Question Answer to Include:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.									
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6a(i)	Hydrolysis	$\text{C}_{15}\text{H}_{31}-\text{C} \begin{array}{l} \text{=O} \\ \text{O}-\text{C}_{20}\text{H}_{29} \end{array} + \text{H}_2\text{O}$ ↓ $\text{C}_{15}\text{H}_{31}-\text{C} \begin{array}{l} \text{=O} \\ \text{OH} \end{array} + \text{H}-\text{O}-\text{C}_{20}\text{H}_{29}$																
6a(ii)	$\text{C}_{20}\text{H}_{29}\text{OH}$ or $\text{C}_{20}\text{H}_{30}\text{O}$																	
6b(i)	Bond breaking by u.v. light	The initiation step forms free radicals by breaking covalent bonds and free radical particles are formed with unpaired electrons e.g. $\text{Cl}-\text{Cl} \rightarrow \text{Cl}^\bullet + \text{Cl}^\bullet$																
6b(ii)	Propagation	<table><tr><td>Step</td><td>Reactants (before Arrow)</td><td>→</td><td>Products (after Arrow)</td></tr><tr><td>Initiation</td><td>No free radicals on Reactant Side</td><td>→</td><td>Free radicals on Product Side</td></tr><tr><td>Propagation</td><td colspan="3">Free Radicals found on both sides of arrow</td></tr><tr><td>Termination</td><td>Free radicals on Reactant Side</td><td>→</td><td>No free radicals on Product Side</td></tr></table>	Step	Reactants (before Arrow)	→	Products (after Arrow)	Initiation	No free radicals on Reactant Side	→	Free radicals on Product Side	Propagation	Free Radicals found on both sides of arrow			Termination	Free radicals on Reactant Side	→	No free radicals on Product Side
Step	Reactants (before Arrow)	→	Products (after Arrow)															
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Termination	Free radicals on Reactant Side	→	No free radicals on Product Side															
6b(iii)	One answer from:	<table><tr><td>Can react with free radicals to form stable molecules</td><td>Donates electrons</td><td>Acts as a reducing agent</td><td>Provides electrons to pair with an unpaired electron</td></tr></table>	Can react with free radicals to form stable molecules	Donates electrons	Acts as a reducing agent	Provides electrons to pair with an unpaired electron												
Can react with free radicals to form stable molecules	Donates electrons	Acts as a reducing agent	Provides electrons to pair with an unpaired electron															
6c(i)	$\begin{array}{c} \text{O} \quad \text{H} \\ \parallel \quad \\ -\text{C}-\text{N}- \end{array}$	CH₃(CH₂)₁₄-C(=O)-NH-(CH₂)₄-C(=O)-NH-CH(CH₃)-CH(OH)-C(=O)-NH-CH(CH₃)-CH(OH)-C(=O)-NH-(CH₂)₄-C(=O)-NH-CH(CH₂OH)-C(=O)-OH																
6c(ii)	CH₃(CH₂)₁₄-C(=O)-NH-CH(CH₃)-C(=O)-OH	Palmitic acid (no -NH₂ group) Sidegroup (CH₂)₄NH₂ Sidegroup CH(OH)CH₃ Sidegroup CH(OH)CH₃ Sidegroup (CH₂)₄NH₂ Sidegroup CH₂CH₃																
7a(i)	One answer from:	Structure 1: $\text{H}_2\text{C}=\text{C}(\text{CH}_3)-\text{CH}=\text{CH}_2$ Structure 2: $\text{H}_3\text{C}-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2$																
7a(ii)	Sesquiterpene	Isoprene (2-methylbuta-1,3-diene) has the formula C_5H_8 . Humulene has formula $\text{C}_{15}\text{H}_{24}$ and is formed when three C_5H_8 units join together.																
7b(i)	5.345	1 flu vaccine ↔ 10.69 mg squalene 500000 flu vaccines ↔ 10.69 mg squalene × 500000/1 = 5345000 mg squalene = 5345g squalene = 5.345 kg squalene																
7b(ii)	6	1 mol of H_2 will add across 1 mol $\text{C}=\text{C}$ double bonds 6 mol of H_2 will add across 6 mol $\text{C}=\text{C}$ double bonds 1 mol of Squalene contains 6 mol $\text{C}=\text{C}$ double bonds ∴ 1 mol of squalene will react with 6 mol of H_2																

7c(i)	Addition or Hydration	Addition Reactions have a molecule adding across a C=C double bond or a C≡C triple bond. If the molecule adding across the C=C double bond is water then the reaction can also be described as hydration.												
7c(ii)	Terpineol is a tertiary alcohol	Hot copper (II) oxide oxidises primary alcohols into carboxylic acids and oxidises secondary alcohols into ketones. Tertiary alcohols do not oxidise.												
8a	286	<table border="1"><thead><tr><th colspan="2">Bond Breaking Steps</th><th colspan="2">Bond Forming Steps</th></tr></thead><tbody><tr><td>6x C-H bonds 1x C-C bond</td><td>6x 412kJ = 2472kJ 1x 348kJ = 348kJ</td><td>1x C≡C bonds 2x C-H bonds 2x H-H bonds</td><td>1x 838kJ = 838kJ 2x 412kJ = 824kJ 2x 436kJ = 872kJ</td></tr><tr><td>Total bond breaking</td><td>= 2820kJ</td><td>Total bond Forming</td><td>= 2534kJ</td></tr></tbody></table> $\Delta H = \Sigma \text{Bond enthalpies for bonds broken} - \Sigma \text{Bond enthalpies for bonds formed}$ $\Delta H = 2820 - 2534$ $\Delta H = +286 \text{ kJ mol}^{-1}$	Bond Breaking Steps		Bond Forming Steps		6x C-H bonds 1x C-C bond	6x 412kJ = 2472kJ 1x 348kJ = 348kJ	1x C≡C bonds 2x C-H bonds 2x H-H bonds	1x 838kJ = 838kJ 2x 412kJ = 824kJ 2x 436kJ = 872kJ	Total bond breaking	= 2820kJ	Total bond Forming	= 2534kJ
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Total bond breaking	= 2820kJ	Total bond Forming	= 2534kJ											
8b	185	① C + O₂ → CO₂ ΔH=-394 kJ mol⁻¹ ② H₂ + ½O₂ → H₂O ΔH=-286 kJ mol⁻¹ ③ C₃H₄ + 4O₂ → 3CO₂ + 2H₂O ΔH=-1939 kJ mol⁻¹ ①x3 3C + 3O₂ → 3CO₂ ΔH=-1182 kJ ②x2 2H₂ + O₂ → 2H₂O ΔH=-572 kJ ③x-1 3CO₂ + 2H₂O → C₃H₄ + 4O₂ ΔH=+1939 kJ Add ①'+②'+③' 3C + 2H₂ → C₃H₄ ΔH=+185 kJ mol⁻¹												
8c(i)	48475	gfm C ₃ H ₄ = 40g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{1000\text{g}}{40} = 25 \text{ mol}$ $1\text{mol} \longleftrightarrow -1939 \text{ kJ}$ $25\text{mol} \longleftrightarrow -1939\text{kJ} \times \frac{25}{1}$ $= 48475 \text{ kJ}$												
8c(ii)	13.76	gfm C ₃ H ₄ = 40g $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{1\text{g}}{40\text{g mol}^{-1}} = 0.025 \text{ mol}$ $\text{C}_3\text{H}_{4(\text{g})} + 4\text{O}_{2(\text{g})} \longrightarrow 3\text{CO}_{2(\text{g})} + 2\text{H}_2\text{O}_{(\text{l})}$ 1mol 0.025mol gfm O₂ = 32g 4mol 0.1mol $\text{mass} = \text{no. of mol} \times \text{gfm} = 0.1 \times 32 = 3.2\text{g}$ mass of air = 4.3 mass of air = 4.3 mass of air = 13.76g x mass of oxygen x 3.2												
8c(iii)	Methanol and ethanol contain oxygen in their structure	Methanol and ethanol are alcohols which contain the -OH hydroxyl group. This oxygen inside the molecule means less oxygen is required from air to burn the structure fully. Ethane and propane are alkanes and these hydrocarbons have no oxygen in their structure.												
9a(i)	Any two from:	<table><tr><td>recycle (waste) gases</td><td>use catalyst</td><td>low/reduce energy requirements</td></tr><tr><td>reactors are run at lower temperatures</td><td>inexpensive feedstocks</td><td>selling/using by-products</td></tr></table>	recycle (waste) gases	use catalyst	low/reduce energy requirements	reactors are run at lower temperatures	inexpensive feedstocks	selling/using by-products						
recycle (waste) gases	use catalyst	low/reduce energy requirements												
reactors are run at lower temperatures	inexpensive feedstocks	selling/using by-products												
9a(ii)	Distillation	Distillation separates chemicals with different boiling points. Ethane-1,2-diol has a higher boiling point due to hydrogen bonding by its two hydroxyl -OH groups.												
9b	Answer to include:	<table><tr><td>1st Mark:</td><td>Propan-1-ol has one hydroxyl -OH group Ethane-1,2-diol has two hydroxyl -OH group</td></tr><tr><td>2nd Mark:</td><td>Stronger/more hydrogen bonding in ethane-1,2-diol than propan-1-ol</td></tr></table>	1 st Mark:	Propan-1-ol has one hydroxyl -OH group Ethane-1,2-diol has two hydroxyl -OH group	2 nd Mark:	Stronger/more hydrogen bonding in ethane-1,2-diol than propan-1-ol								
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9c	One diagram from:	OH H₃C—C—CH₃ OH (propane-2,2-diol) OH H₃C—CH₂—CH OH (propane-1,1-diol) OH OH H₃C—CH—CH₂ (propane-1,2-diol) OH OH H₂C—CH₂—CH₂ (propane-1,3-diol)																														
9d(i)	Answer to include:	1st Mark: Use pipette to measure 20cm³ of ethanol 2nd Mark: Fill up 100cm³ volumetric/standard flask up to mark with deionised water																														
9d(ii)	157.5	1kg animal ↔ 5cm³ ethanol solution 3.5kg animal ↔ 5cm³ × 3.5/1 ethanol solution = 17.5cm³ 1 dose for 3.5kg animal ↔ 17.5cm³ ethanol solution 9 doses for 3.5kg animal ↔ 17.5cm³ × 9/1 = 157.5cm³																														
9d(iii) PART A	One answer from:	O O H—O—C—C—O—H O O H—O—C—C—H O O H—C—C—H H OH H—C—C—H																														
9d(iii) PART B	HOCH ₂ COONa or NaC ₂ H ₃ O ₃	HOCH ₂ COOH + NaOH → HOCH ₂ COONa + H ₂ O																														
10	Open Question Answer to Include:	<table><tr><td>3 mark answer</td><td>2 mark answer</td><td>1 mark answer</td></tr><tr><td>Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																								
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11a	One from:	Weighing By Difference Set the balance to zero, place weighing bottle on balance. Record empty mass. Add 50.0g of salt to weighing bottle on balance and record the new mass. The difference in masses is the mass of the substance in the bottle. Tare Method Place weighing boat/bottle on the balance. Press TARE button on balance. When reading is a zero, add 50.0g of salt and read the mass on the balance.																														
11b	2I ⁻ → I ₂ + 2e ⁻	Redox equation: 2I⁻ + Br₂ → I₂ + 2Br⁻ Reduction step: Br₂ + 2e⁻ → + 2Br⁻ Oxidation step: 2I⁻ → I₂ + 2e⁻																														
11c(i)	9.5	Average volume = $\frac{9.4 + 9.6}{2} = \frac{19.0}{2} = 9.5\text{cm}^3$																														
11c(ii)	4.75×10 ⁻⁶ or 0.00000475	no. of mol = volume × concentration = 0.0095litres × 0.0010mol l⁻¹ = 9.5×10⁻⁶mol I₂ + 2Na₂S₂O₃ → 2NaI + Na₂S₄O₆ 1mol 2mol 4.75×10⁻⁶mol 9.5×10⁻⁶mol																														
12a(i)	1st Mark More chlorine atoms increases germ-killing power 2nd Mark Longer carbon chain increases germ-killing power	<table><tr><td>Compound</td><td>Phenol</td><td>2-chlorophenol</td><td>2,4-dichlorophenol</td><td>2,4,6-trichlorophenol</td></tr><tr><td>No. of Chlorine atoms</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Germ-killing Power</td><td>1.0</td><td>3.6</td><td>13.0</td><td>23.0</td></tr></table> Conclusion: Increasing the number of chlorines increases the germ-killing power <table><tr><td>Compound</td><td>Phenol</td><td>4-methylphenol</td><td>4-ethylphenol</td><td>4-propylphenol</td></tr><tr><td>Carbon Chain Length</td><td>0</td><td></td><td>2</td><td>3</td></tr><tr><td>Germ-killing Power</td><td>1.0</td><td>2.5</td><td>7.5</td><td>20.0</td></tr></table> Conclusion: Increasing the carbon chain length increases the germ-killing power	Compound	Phenol	2-chlorophenol	2,4-dichlorophenol	2,4,6-trichlorophenol	No. of Chlorine atoms	0	1	2	3	Germ-killing Power	1.0	3.6	13.0	23.0	Compound	Phenol	4-methylphenol	4-ethylphenol	4-propylphenol	Carbon Chain Length	0		2	3	Germ-killing Power	1.0	2.5	7.5	20.0
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12a(ii)	2-chloro-4,5-dimethylphenol	2-chloro-4,5-dimethylphenol Chlorine on C₂ of ring (anti-clockwise numbering) -CH₃ groups on C₄ & C₅ (anti-clockwise numbering) 6 carbon ring with -OH group on C₁																														

12b(i)	126.9	$\text{C}_6\text{H}_6 + \text{H}_2\text{SO}_4 + 2\text{NaOH} \longrightarrow \text{C}_6\text{H}_5\text{OH} + \text{Na}_2\text{SO}_3 + 2\text{H}_2\text{O}$ 1mol 78g 117g 117kg 1mol 94g $94\text{g} \times \frac{117}{78}$ = 141g 141kg $\% \text{Yield} = \frac{\text{Actual}}{\text{Theoretical}} \times 100 \quad \therefore \text{Actual} = \frac{\% \text{Yield} \times \text{Theoretical}}{100} = \frac{90 \times 141\text{kg}}{100} = 126.9\text{kg}$				
12b(ii)	propanone	Cumene hydroperoxide [ring]-C₃H₇O₂ $\longrightarrow$ $\longrightarrow$ Phenol [ring]-OH + + X C₃H₆O C₃H₆O has two possible structures. <table style="width: 100%; text-align: center;"> <tr> <th style="width: 50%;">Propanone</th> <th style="width: 50%;">propanal</th> </tr> <tr> <td> $\begin{array}{ccccc} & \text{H} & \text{O} & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} - \text{H} \\ & & & & \\ & \text{H} & & \text{H} & \end{array}$ </td> <td> $\begin{array}{ccccc} & \text{H} & \text{H} & & \text{O} \\ & & & & // \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} \\ & & & & \backslash \\ & \text{H} & \text{H} & & \text{H} \end{array}$ </td> </tr> </table> The oxygen atoms in cumene hydroperoxide are attached to C₂ of three carbon chain Conclusion: X must be propanone	Propanone	propanal	$\begin{array}{ccccc} & \text{H} & \text{O} & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} - \text{H} \\ & & & & \\ & \text{H} & & \text{H} & \end{array}$	$\begin{array}{ccccc} & \text{H} & \text{H} & & \text{O} \\ & & & & // \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} \\ & & & & \backslash \\ & \text{H} & \text{H} & & \text{H} \end{array}$
Propanone	propanal					
$\begin{array}{ccccc} & \text{H} & \text{O} & \text{H} & \\ & & & & \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} - \text{H} \\ & & & & \\ & \text{H} & & \text{H} & \end{array}$	$\begin{array}{ccccc} & \text{H} & \text{H} & & \text{O} \\ & & & & // \\ \text{H} & - \text{C} & - & \text{C} & - \text{C} \\ & & & & \backslash \\ & \text{H} & \text{H} & & \text{H} \end{array}$					

2018 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	B	2	70	24	3	0	0
2	A	91	4	3	2	0	0
3	D	9	12	8	70	0	0
4	C	10	17	57	16	1	0
5	D	21	6	7	65	0	0
6	C	2	1	77	20	0	0
7	A	39	8	8	44	0	0
8	D	23	6	6	64	0	0
9	A	69	9	12	9	1	0
10	B	5	93	1	1	0	0
11	D	8	11	37	43	1	0
12	D	26	8	32	34	1	0
13	A	48	22	9	19	1	0
14	A	52	14	25	8	1	0
15	D	15	26	15	43	1	0
16	C	11	7	67	15	1	0
17	B	9	75	14	3	0	0
18	C	24	15	43	18	1	0
19	B	23	49	10	18	1	0
20	B	13	59	13	15	0	0